

M(6) generation

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1 Partitions

Given all the sources for $M(1)$, $M(2)$, $M(3)$, $M(4)$, and $M(5)$

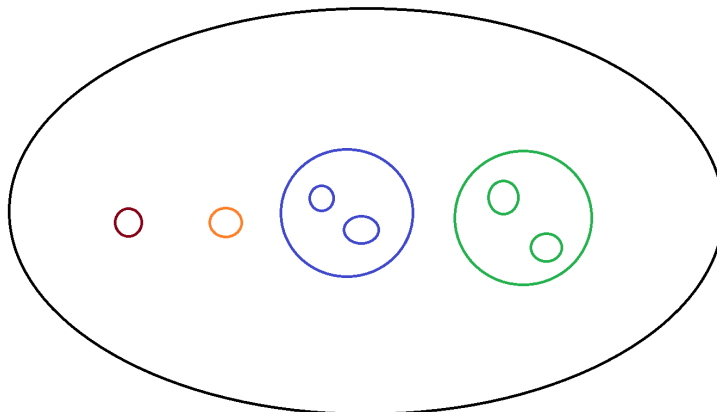
We can find $M(6)$

For each source in $M(1)$, we can append any sources from any partition of 5 with more than 2 pieces.

The partitions of 5 with more than 2 pieces are:

$1 + 1 + 1 + 1 + 1$
 $1 + 1 + 1 + 2$
 $1 + 1 + 3$
 $1 + 2 + 2$
 $1 + 4$
 $2 + 3$

For example given the atom from $M(1)$ we can append sources from $M(1)$ and $M(2)$ and $M(2)$, since $1 + 2 + 2$ is a partition of 5 with more than 2 pieces. Figure 1 gives an example of the concatenation of these four sources from $M(1)$, $M(1)$, $M(2)$, and $M(2)$.



The following pages will show the generation of the sources of $M(6)$

For reference, the partitions of 2, 3, 4, and 5 with more than 2 pieces are given now.

2:

$$1 + 1$$

3:

$$1 + 1 + 1$$

$$1 + 2$$

$$3$$

4:

$$1 + 1 + 1 + 1$$

$$1 + 1 + 2$$

$$1 + 3$$

$$2 + 2$$

5:

$$1 + 1 + 1 + 1 + 1$$

$$1 + 1 + 1 + 2$$

$$1 + 1 + 3$$

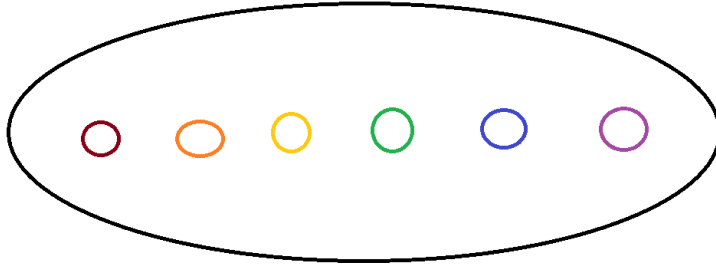
$$1 + 2 + 2$$

$$1 + 4$$

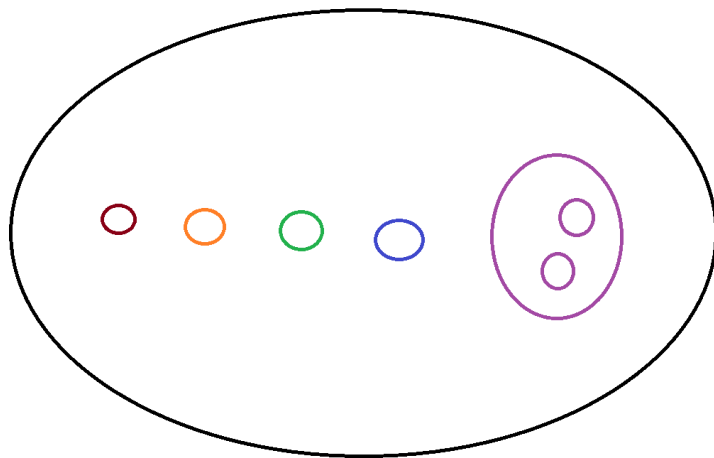
$$2 + 3$$

We will have to check for repeats.

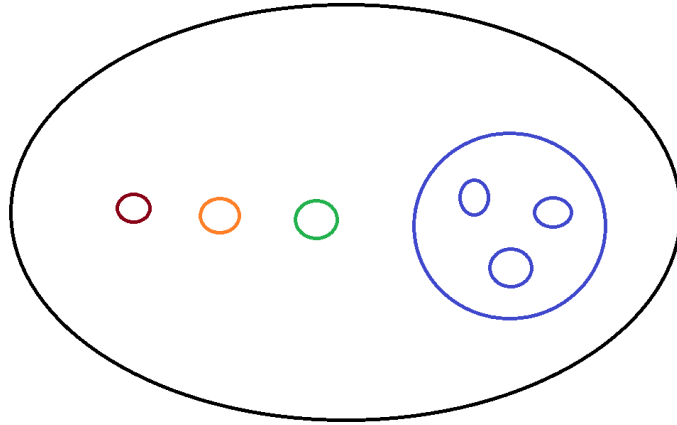
$$M(1) + [M(1) + M(1) + M(1) + M(1) + M(1)]$$



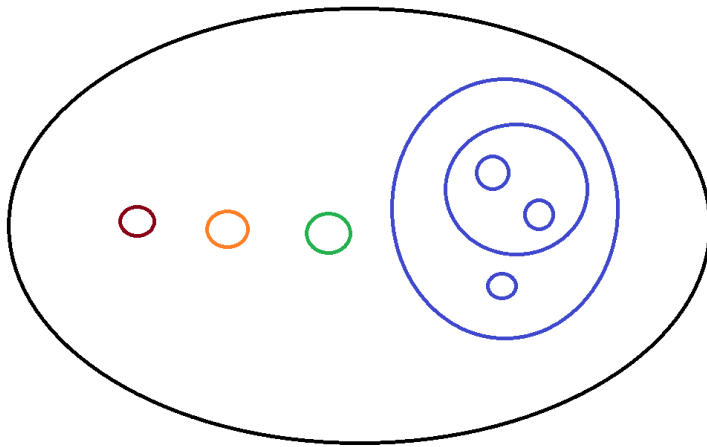
$$M(1) + [M(1) + M(1) + M(1) + M(2)]$$



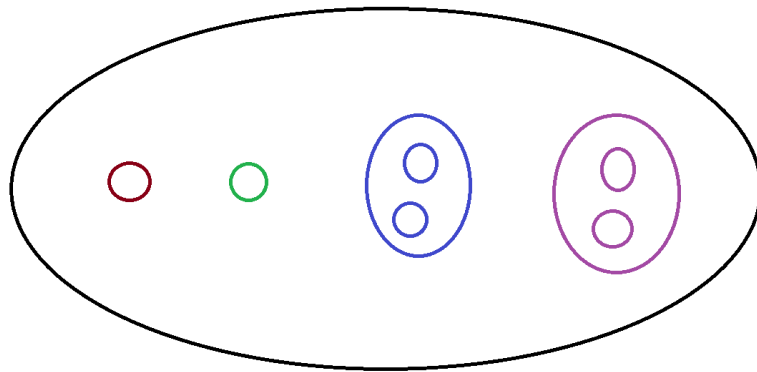
$$M(1) + [M(1) + M(1) + M(3)]$$



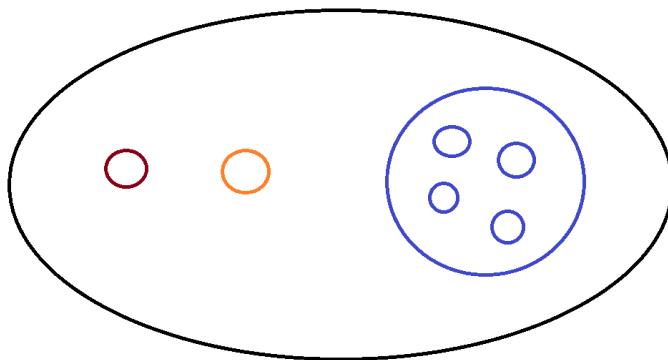
$$M(1) + [M(1) + M(1) + M(3)]$$



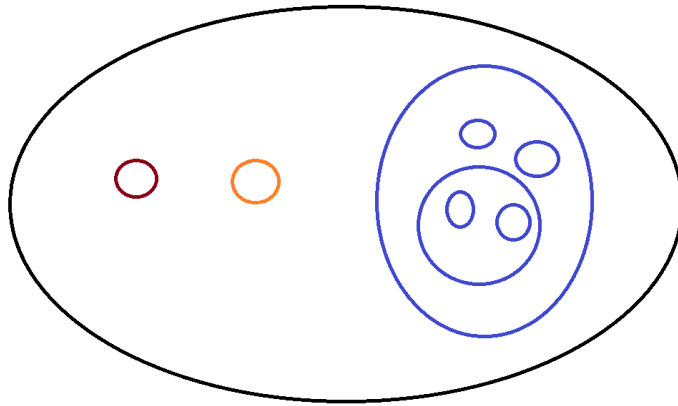
$$M(1) + [M(1) + M(2) + M(2)]$$



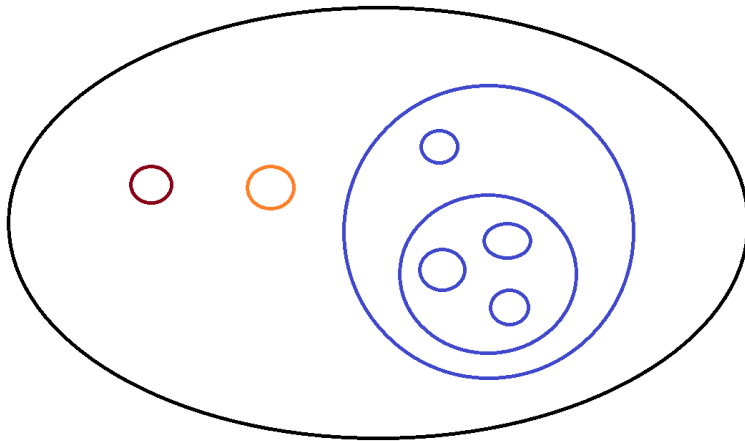
$$M(1) + [M(1) + M(4)]$$



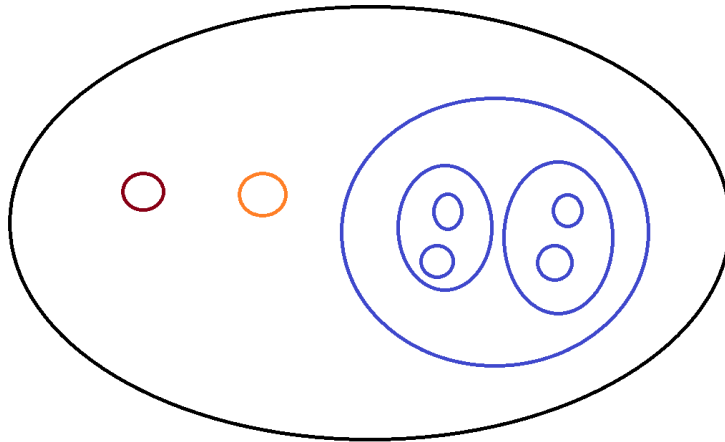
$$M(1) + [M(1) + M(4)]$$



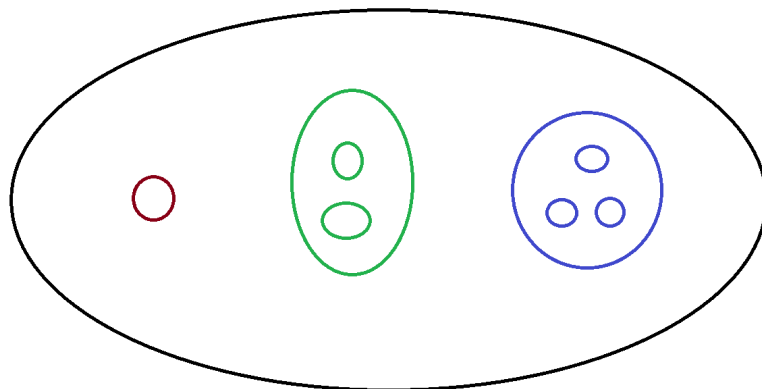
$$M(1) + [M(1) + M(4)]$$



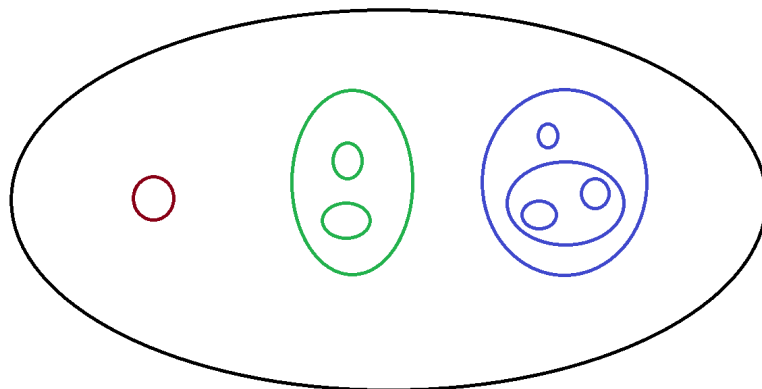
$$M(1) + [M(1) + M(4)]$$



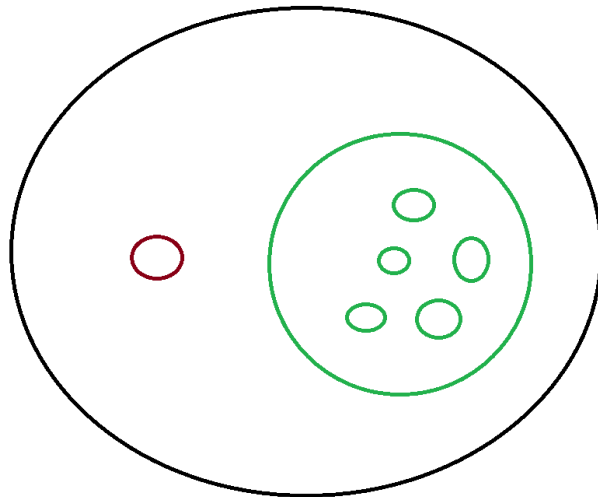
$$M(1) + [M(2) + M(3)]$$



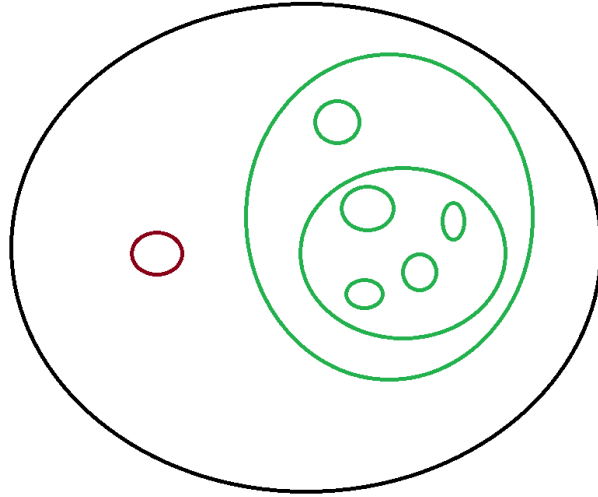
$$M(1) + [M(2) + M(3)]$$



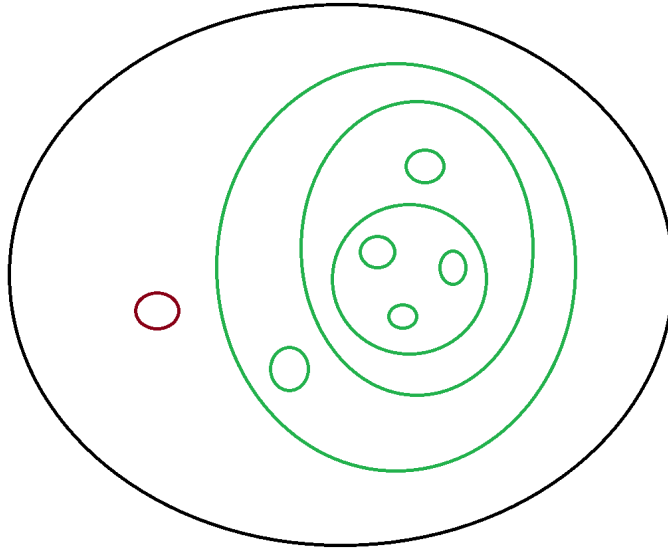
$$M(1) + [M(5)]$$



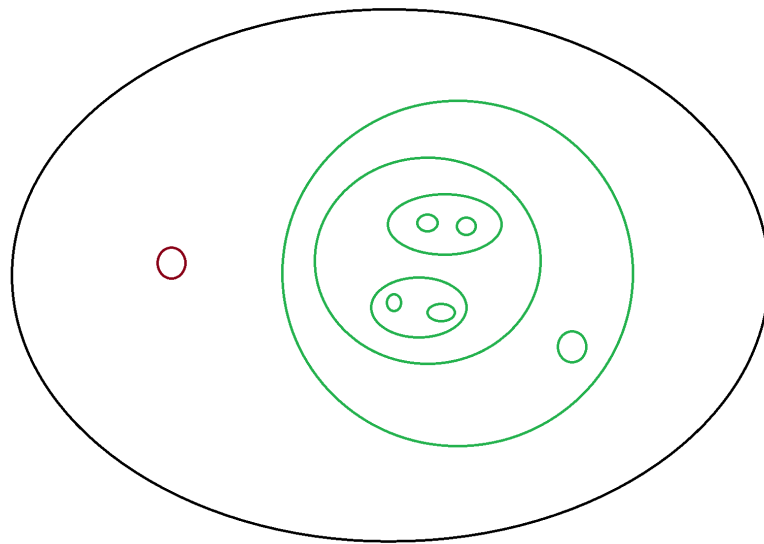
$$M(1) + [M(5)]$$



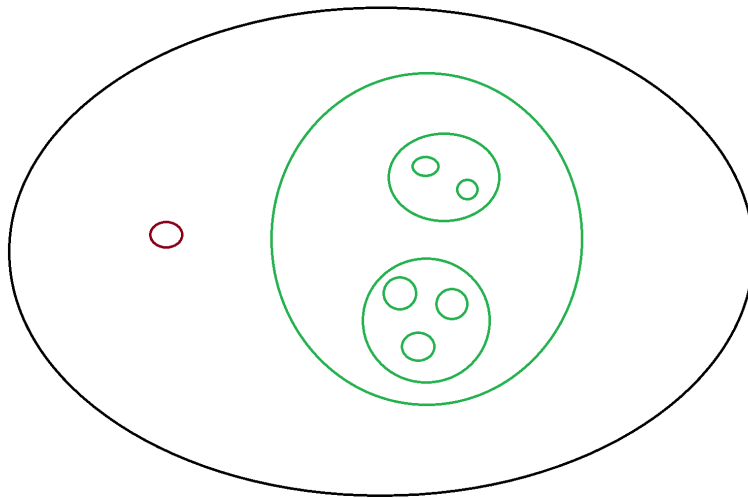
$$M(1) + [M(5)]$$



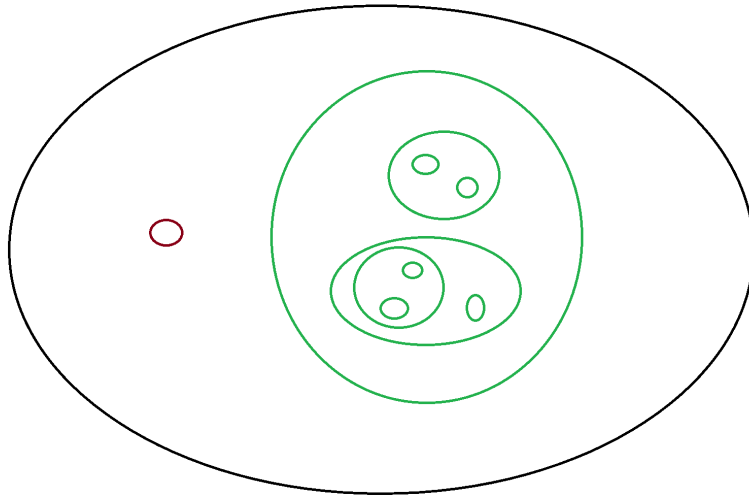
$$M(1) + [M(5)]$$



$$M(1) + [M(5)]$$



$$M(1) + [M(5)]$$



$$M(2) + [M(1) + M(1) + M(1) + M(1)]$$

Repeat of above.

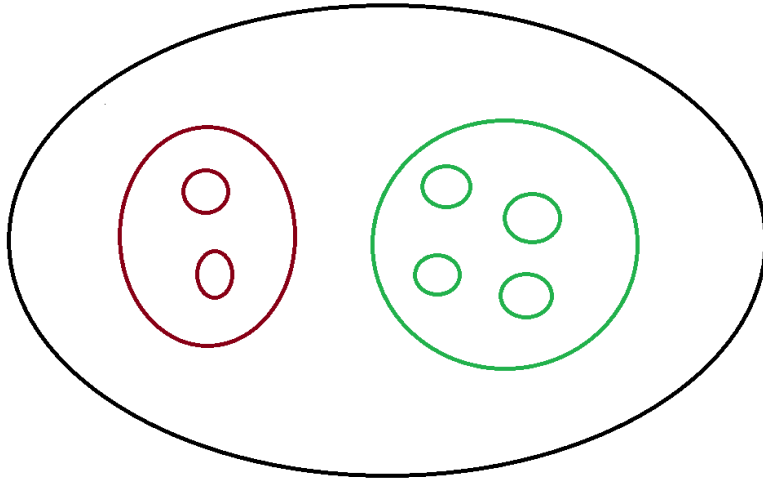
$$M(2) + [M(1) + M(1) + M(2)]$$

Repeat of above.

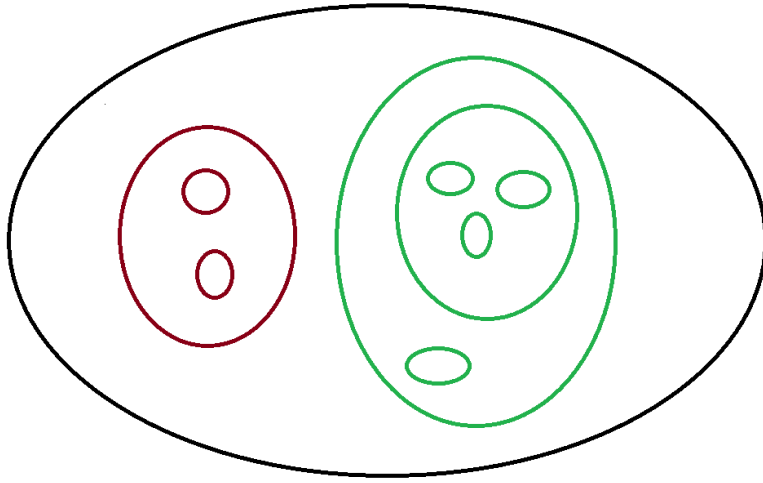
$$M(2) + [M(1) + M(3)]$$

Repeat of above.

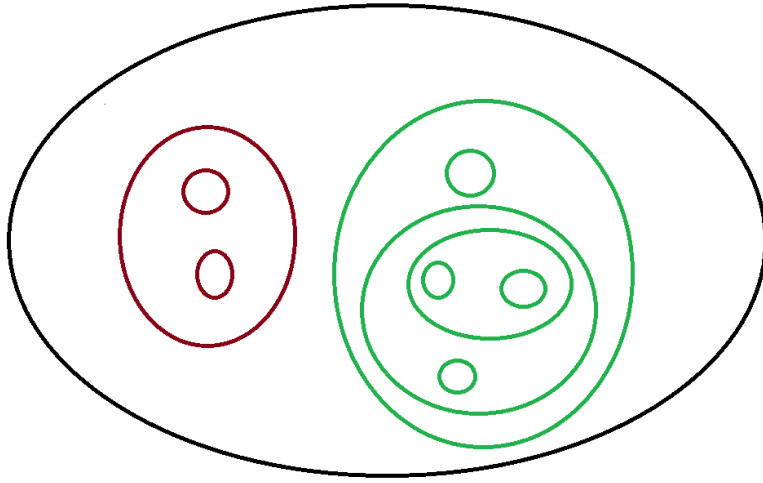
$$M(2) + [M(4)]$$



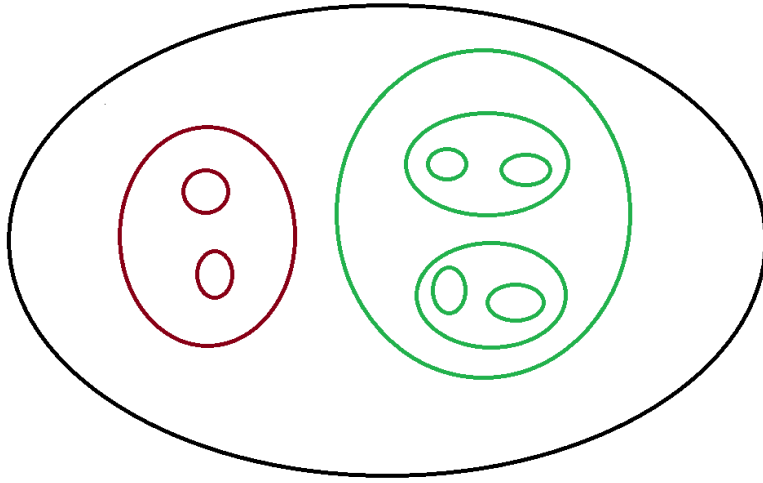
$$M(2) + [M(4)]$$



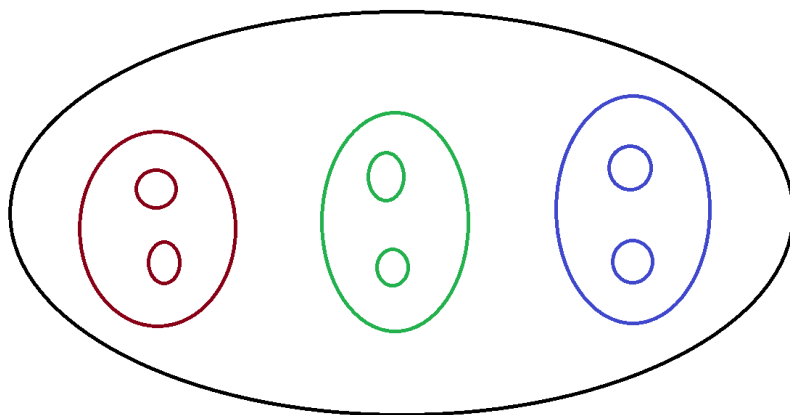
$$M(2) + [M(4)]$$



$$M(2) + [M(4)]$$



$$M(2) + [M(2) + M(2)]$$



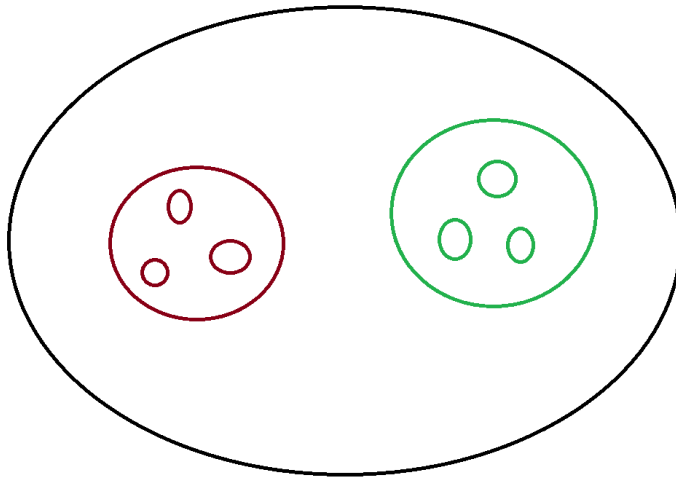
$$M(3) + [M(1) + M(1) + M(1)]$$

Repeat of above.

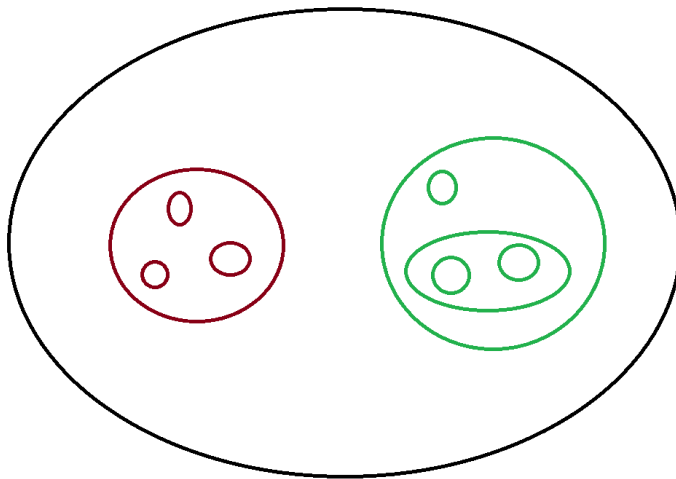
$$M(3) + [M(1) + M(2)]$$

Repeat of above.

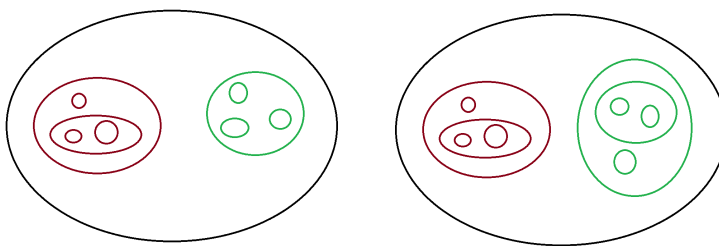
$$M(3) + [M(3)]$$



$$M(3) + [M(3)]$$



Repeats:



$$M(4) + [M(1) + M(1)]$$

Repeat of above.

$$M(4) + [M(2)]$$

Repeat of above.

$$M(5) + [M(1)]$$

Repeat of above.